

Q1. Consider the following dataset representing the number of students enrolled in different courses:

`courses = ["BCA", "BBA", "B.Com", "B.Sc", "BA"]`

`students = [60, 45, 75, 50, 35]`

Create a Bar Chart to visualize the number of students enrolled in each course. Use different bar colors, add an appropriate title, and label both axes.

[10 Marks]

Q2. Implement a KNN classifier on the given Student dataset. Preprocess the data and classify students into different performance categories. Evaluate the model using Accuracy and Confusion Matrix.

[20 Marks]

OR

Q2. Write a Python program to perform market basket analysis on the Groceries dataset using the Apriori algorithm. Read and explore the dataset, preprocess the data by handling missing values, convert categorical transaction data into a suitable numeric format, and generate frequent itemsets and association rules. Repeat the analysis using minimum support values 0.3 and 0.4.

[20 Marks]

S.No.	Items Purchased
1	Basmati Rice, Toor Dal, Turmeric Powder, Cooking Oil
2	Atta, Sugar, Tea, Milk
3	Rice, Moong Dal, Salt, Cooking Oil
4	Wheat Flour, Chickpeas, Turmeric Powder, Cumin Seeds
5	Poha, Peanuts, Onions, Cooking Oil
6	Basmati Rice, Paneer, Garam Masala, Tomatoes
7	Atta, Potatoes, Onions, Cooking Oil

Q3. Viva

[5 Marks]